Set
Wed 9 Sep 2026

Homework 3 — Functions

Session 3 · Domain, codomain, and undoing

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

BEFORE YOU START No question here is answered by a verdict alone. "Injective" is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is. To *disprove* injectivity or surjectivity you need one concrete witness — to *prove* either, you need an argument covering every element.

Part A · Mechanics

1 IMAGES AND PREIMAGES

Let $f : \mathbb{Z} \to \mathbb{Z}$ be given by $f(n) = 3n - 2$. Compute $f(\{-1, 0, 2\})$, $f^{-1}(\{7\})$ and $f^{-1}(\{5\})$. Explain in one sentence why the third answer is what it is.

2 IMAGE AGAINST CODOMAIN

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 4x + 7$. State the image of f and prove your answer. Is f surjective? Justify.

3 UNIT CIRCLE

Derive each value from the unit circle. For each, state the quadrant and the reference angle before you give the number. A recalled value without that working is not a derivation.

a. $\cos \dfrac{2\pi}{3}$

b. $\sin \dfrac{3\pi}{4}$

c. $\cos \dfrac{7\pi}{6}$

d. $\sin \dfrac{5\pi}{3}$

e. $\tan \dfrac{3\pi}{4}$

Part B · Proof

4 CLASSIFICATION

For each function decide whether it is injective, surjective, both or neither. Prove each positive claim; disprove each negative one with an explicit witness.

a. $f : \mathbb{R} \to \mathbb{R},\ f(x) = 5 - 3x$

b. $f : \mathbb{Z} \to \mathbb{Z},\ f(n) = n^2 - n$

c. $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^3 - 3x$

d. $f : \mathbb{N} \to \mathbb{N},\ f(n) = n^2$

e. $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\},\ f(x) = \dfrac{1}{x}$

f. $f : \mathbb{R} \to [-1, 1],\ f(x) = \sin x$

g. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},\ f(a, b) = 2a + 4b$

h. $S = \{1, 2, 3\},\ f : \mathcal{P}(S) \to \mathcal{P}(S),\ f(X) = S \setminus X$

5 COMPOSITION

Let $f : A \to B$ and $g : B \to C$.

a. Prove that if $g \circ f$ is surjective, then g is surjective.

b. Show by explicit example that f need *not* be surjective under the same hypothesis.

c. In one sentence, say why (a) is about g and not f, given that in class we proved $g \circ f$ injective forces f injective.

6 RESTRICTION

Take the rule $x \mapsto x^2 - 4x + 7$ from question 2. Choose a domain $A \subseteq \mathbb{R}$ and a codomain B making $f : A \to B$ a **bijection**, and prove both halves.

Part C · Communication

7 FIND THE ERROR

The following was produced by a chatbot. It contains **three** distinct errors. Find all three; for each, say what is wrong and give a witness where one is needed.

> Claim. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{x^2+1}$, is a bijection, and its inverse is $f^{-1}(x) = \dfrac{x^2+1}{x}$.
>
> "Proof." f is injective because it is a ratio of polynomials with no repeated factors. It is surjective because its domain and codomain are both $\mathbb{R}$. Inverting a fraction gives the inverse function.

8 REWRITE PRECISELY

Each sentence is sloppy or false as written. Rewrite each so that it is unambiguous and true.

a. "x^2 is not injective."

b. "Every function has an inverse — you just swap x and y."

c. "$\sin^{-1}(x)$ means $\dfrac{1}{\sin x}$."